

Spazi di Hilbert

Def. Siano H uno sp. vettoriale.

Un **PRODOTTO SCALARE** su H è un'applicazione

$$(\cdot, \cdot) : H \times H \longrightarrow \mathbb{R} \quad \text{tale che}$$

$$(i) \quad (x, x) \geq 0 \quad \forall x \in H \quad \text{con} \quad = \iff x=0$$

$$(ii) \quad (x, y) = (y, x) \quad \forall x, y$$

$$(iii) \quad (\alpha_1 x_1 + \alpha_2 x_2, y) = \alpha_1 (x_1, y) + \alpha_2 (x_2, y) \\ \forall x_1, x_2, y \in H, \quad \forall \alpha_1, \alpha_2 \in \mathbb{R}$$

$$\underline{\text{Om.}} \quad (x, \alpha_1 y_1 + \alpha_2 y_2) \stackrel{(i)}{=} \alpha_1 (x, y_1) + \alpha_2 (x, y_2) \\ \text{" (ii)} \quad \text{" (ii)}$$

$$(\alpha_1 y_1 + \alpha_2 y_2, x) \stackrel{(iii)}{=} \alpha_1 (y_1, x) + \alpha_2 (y_2, x)$$

Esempi

$$\bullet \quad H = \mathbb{R}^n \quad (x, y) = (x \cdot y = \langle x, y \rangle) \stackrel{\text{def}}{=} \sum_{i=1}^n x_i y_i \\ (x_1, \dots, x_n) \quad (y_1, \dots, y_n)$$

$$\bullet \quad H = L^2(\Omega) \quad (f, g)_{L^2} \stackrel{\text{def}}{=} \int_{\Omega} f g \in \mathbb{R}$$

$$\bullet \quad H = H^1(\Omega) \\ = W^{1,2}(\Omega) \quad (f, g)_{H^1} \stackrel{\text{def}}{=} \int_{\Omega} f g + \nabla f \cdot \nabla g \quad \text{Stöckler}$$

$$\begin{aligned}
 n=1 & \quad (f, g)_{H^1} = \int_{\Omega} f g + f' g' \\
 n \geq 1 & \quad (f, g)_{H^1} = \int_{\Omega} f g + \sum_{i=1}^n \frac{\partial f}{\partial x_i} \cdot \frac{\partial g}{\partial x_i} \quad \left\{ \begin{array}{l} \in \mathbb{R} \\ \uparrow \\ \text{Hölder} \end{array} \right.
 \end{aligned}$$

Def. Sia H sp. vett., con un prodotto scalare $(,)$

Allora si dice NORMA INDOTTA del prodotto scalare

$$\|x\| := \sqrt{(x, x)}$$

Esempio

- $H = \mathbb{R}^n$ $\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$
- $H = L^2(\Omega)$ $\|f\|_2 = \sqrt{\int_{\Omega} f^2} = \left(\int_{\Omega} f^2 \right)^{1/2}$
- $H = H^1(\Omega)$ $\|f\|_{H^1} = \left(\int_{\Omega} f^2 + \underbrace{\nabla f \cdot \nabla f}_{|\nabla f|^2} \right)^{1/2}$

Q. È una norma (davvero)?

$$1) \quad \|x\| \geq 0 \quad \text{con} \quad \|x\| = 0 \iff x = 0$$
$$\sqrt{\langle x, x \rangle}$$

vero per (i) del prodotto scalare

$$2) \quad \|\lambda x\| = |\lambda| \|x\|$$
$$\|\lambda x\|^2 = |\lambda|^2 \|x\|^2$$

vero perché

$$\|\lambda x\|^2 = \langle \lambda x, \lambda x \rangle \stackrel{(iii)}{=} \lambda^2 \|x\|^2 = |\lambda|^2 \|x\|^2$$

$$3) \quad \|x+y\| \stackrel{?}{\leq} \|x\| + \|y\|$$

$$\|x+y\|^2 \leq \|x\|^2 + \|y\|^2 + 2\|x\|\|y\|$$

In termini del prodotto scalare:

$$\langle x+y, x+y \rangle \leq \langle x, x \rangle + \langle y, y \rangle + 2\sqrt{\langle x, x \rangle} \sqrt{\langle y, y \rangle}$$

$$\langle x, x \rangle + \langle y, y \rangle$$

$$+ 2\langle x, y \rangle$$

$$\langle x, y \rangle \leq \sqrt{\langle x, x \rangle} \sqrt{\langle y, y \rangle}$$

Cauchy-Schwarz

Sei (H) sp. vet. munito di prodotto scalare

$$|(x, y)| \leq \|x\| \cdot \|y\| \quad \forall x, y \in H$$

$$\text{con} = (\Leftrightarrow) \quad x = \lambda y \quad \text{per qualche } \lambda \in \mathbb{R}$$

Dim. $\forall t \in \mathbb{R}$

$$0 \leq \underset{\substack{\uparrow \\ (i)}}{(x - ty, x - ty)} = \|x\|^2 - 2t(x, y) + t^2 \|y\|^2$$

$$\Rightarrow \Delta = 4(x, y)^2 - 4\|x\|^2 \|y\|^2 \leq 0$$

$$(x, y)^2 \leq \|x\|^2 \|y\|^2$$

$$|(x, y)| \leq \|x\| \|y\| \quad \text{C. S.}$$

$$\begin{aligned} (\Leftrightarrow) \quad x = \lambda y & \quad \begin{aligned} (x, y) &= (\lambda y, y) = \lambda (y, y) = \lambda \|y\|^2 \\ |(x, y)| &= |\lambda| \|y\|^2 \\ \|x\| \cdot \|y\| &= \underbrace{\|\lambda y\|}_{\substack{= |\lambda| \|y\|}} \cdot \|y\| = |\lambda| \|y\|^2 \end{aligned} \end{aligned}$$

$$(\Rightarrow) \quad \text{Se vale in C.S.}, \Delta = 0$$

$$\Rightarrow \exists \lambda \in \mathbb{R} : (x - \lambda y, x - \lambda y) = 0$$

$$\underset{(i)}{\Rightarrow} \quad x - \lambda y = 0 \Rightarrow x = \lambda y. \quad \square$$

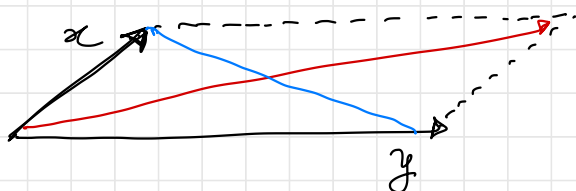
Legge del parallelogramma

Se H è uno sp. vett. munito di prodotto scalare $(\cdot, \cdot)$, e sia $\|\cdot\|$ la norma indotta dal pr. scalare.

Allora: $\forall x, y \in H$

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

In $\mathbb{R}^2$:



Dim.

$$\begin{aligned} \|x+y\|^2 + \|x-y\|^2 &= (x+y, x+y) + (x-y, x-y) = \\ &= (x, x) + (y, y) + 2(x, y) \\ &\quad + (x, x) + (y, y) - 2(x, y) \\ &= 2\|x\|^2 + 2\|y\|^2 \end{aligned} \quad \square$$

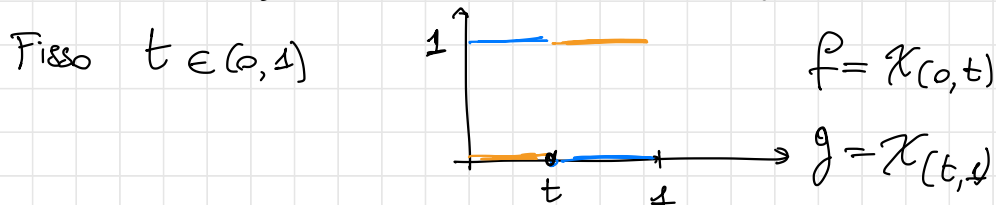
$\mathbb{R}^n, \|\cdot\|_p$ con $p \neq 2$
 $L^p(\Omega), \|\cdot\|_p$ con $p \neq 2$
 $W^{1,p}(\Omega), \|\cdot\|_p$ con $p \neq 2$

NON provengono da un prodotto scalare!

Esempio $H = L^p(\Omega)$ con $p \neq 2$

$$\Omega = (0, 1) \subseteq \mathbb{R}$$

Trovo f e g tali che id. parallelogramma NON vale.



- $\|f\|_p = \left(\int_0^1 |f|^p \right)^{1/p} = \left(\int_0^t 1 \right)^{1/p} = t^{1/p}$
- $\|g\|_p = \left(\int_0^1 |g|^p \right)^{1/p} = \left(\int_t^1 1 \right)^{1/p} = (1-t)^{1/p}$
- $\|f+g\|_p = \left(\int_0^1 |f+g|^p \right)^{1/p} = \left(\int_0^1 1 \right)^{1/p} = 1$
- $\|f-g\|_p = \left(\int_0^1 |f-g|^p \right)^{1/p} = \left(\int_0^1 1 \right)^{1/p} = 1$

legge del parallelogramma

$$2 \|f\|_p^2 + 2 \|g\|_p^2 = 2 t^{\frac{2}{p}} + 2 (1-t)^{\frac{2}{p}}$$

$\stackrel{?}{=}$

$$\|f+g\|_p^2 + \|f-g\|_p^2 =$$

2

↑
FALSO per $p \neq 2$
per qualche t

Caso particolare $(p=2)$ $2t + 2(1-t) \stackrel{?}{=} 2$ VERO

Def. Sia H è uno sp. vett munito di pr-scalar
 $(,)$, e $\| \cdot \|$ è la norma indotta:

$$\bullet \quad x \perp y \stackrel{\text{def}}{\iff} (x, y) = 0$$

Esempi

- In $\mathbb{R}^n$ $x \perp y \iff x \cdot y = 0$ pr-scalar "usuale"
- In $L^2(\Omega)$ $f \perp g \iff \int_{\Omega} f g = 0$
- In $H^1(\Omega)$ $f \perp g \iff \int_{\Omega} f g + \nabla f \cdot \nabla g = 0$.

• Dato $M \subseteq H$,

$$M^{\perp} := \{ y \in H, \text{ tale che } (y, x) = 0 \quad \forall x \in M \}$$

Esempi

$$\bullet \text{ In } \mathbb{R}^3 \quad M = \langle e_1 \rangle, \quad M^{\perp} = \langle e_2, e_3 \rangle$$

$$\bullet \text{ In } L^2(\Omega) \quad M = \{ c, c \in \mathbb{R} \}$$

$$\uparrow$$

$$|\Omega| < +\infty$$

$$M^{\perp} = \{ f \in L^2(\Omega) \text{ tale che } (f, c) = 0 \quad \forall c \in \mathbb{R} \}$$

$$= \{ f \in L^2(\Omega) : \int_{\Omega} f = 0 \} \quad \int_{\Omega} f = 0$$

Teoremi di proiezione

Teoremi di rappresentazione

Def. Uno **SPAZIO DI HILBERT** è
uno sp. vett. munito di pr. scalare
che è DI BANACH munito della
norma indotta dal prodotto scalare.

Esempi

- $H = \mathbb{R}^n, L^2(\Omega), H^1(\Omega), H_0^1(\Omega)$ Hilbert
- $H = C^0([a, b])$ $(f, g) = \int_a^b f \cdot g$
 $\uparrow$
non Hilbert $\|f\|_2 = \sqrt{\int_a^b f^2}$

Teorema di proiezione su un convesso chiuso

Sia H Hilbert, e sia $K \subseteq H$ convesso chiuso.

$$\begin{cases} K \text{ convesso} \stackrel{\text{def}}{=} \forall x, y \in K, \forall \lambda \in [0, 1] \Rightarrow (1-\lambda)x + \lambda y \in K \\ K \text{ chiuso} \stackrel{\text{def}}{=} \forall \{x_n\} \subseteq K, x_n \xrightarrow[H]{\|x_n - x\| \rightarrow 0} x \Rightarrow x \in K \end{cases}$$

Allora:

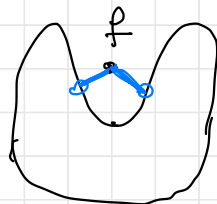
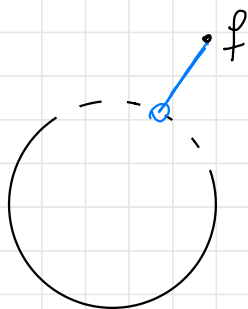
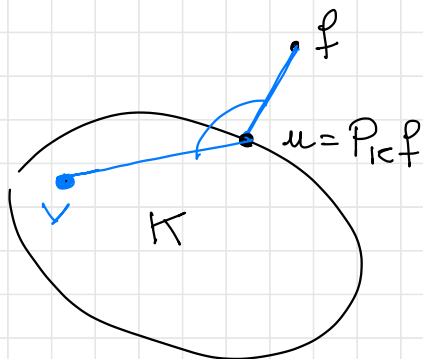
$\forall f \in H$ esiste unico $u \in K$ ($u := P_K f$) tale che

$$\|f - u\| = \min_{v \in K} \|f - v\|$$

$d(f, u) \qquad d(f, v)$

Inoltre u è caratterizzato da

$$u = P_K f \iff \underline{(f - u, v - u) \leq 0} \quad \forall v \in K \quad (1)$$



Corollario (teorema di proiezione su un sottospazio chiuso)

Sia H Hilbert e sia M sottospazio chiuso.

$$\left(\begin{array}{l} M \text{ sottospazio} \Rightarrow M \text{ convesso} \\ \forall x, y \in M, \forall \lambda \in [0, 1] \Rightarrow \lambda x + (1-\lambda)y \in M \end{array} \right)$$

$$\forall f \in H \quad \exists! \quad u = P_M f \text{ tale che}$$

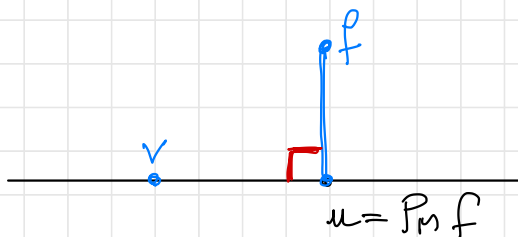
$$\|f - u\| = \min_{v \in M} \|f - v\|$$

Inoltre:

$$u = P_M f \Leftrightarrow (f - u, v) = 0 \quad \forall v \in M \quad (2)$$

$$(f - u) \perp v \quad \forall v \in M$$

$$(f - u \in M^\perp)$$



Verifica $\downarrow$ (1) $\Leftrightarrow$ (2) quando $K=M$ detto pario

($\Leftarrow$) Suppongo $(f-u, v) = 0 \quad \forall v \in M$

Voglio dim. $(f-u, v-u) \leq 0 \quad \forall v \in M$

$$\text{Ho: } (f-u, v-u) = \underbrace{(f-u, v)}_0 - \underbrace{(f-u, u)}_0 = 0$$

($\Rightarrow$) Suppongo $(f-u, v-u) \leq 0 \quad \forall v \in M$.

Voglio dim. $(f-u, w) = 0 \quad \forall w \in M$.

Prendendo $v = u + w \quad (\Rightarrow v-u = w)$.

Per ip. $(f-u, \underbrace{v-u}_w) \leq 0 \quad \forall w \in M$

$$\Rightarrow (f-u, w) = 0 \quad \forall w \in M$$

Osservazione

M sottospazio chiuso $\Rightarrow M^\perp = \{y \in H, (y, x) = 0 \forall x \in M\}$

• $M^\perp$ sottospazio

$$y_1, y_2 \in M^\perp, \alpha_1, \alpha_2 \in \mathbb{R} \Rightarrow \alpha_1 y_1 + \alpha_2 y_2 \in M^\perp$$

$$(\alpha_1 y_1 + \alpha_2 y_2, x) = 0 \quad \forall x \in M \quad ?? \quad \text{e' perche'}$$

$$\alpha_1 \underbrace{(y_1, x)}_0 + \alpha_2 \underbrace{(y_2, x)}_0$$

$$\leftarrow \text{perche' } y_1, y_2 \in M^\perp$$

• $M^\perp$ chiuso. $y_m \in M^\perp, y_m \xrightarrow{H} y \Rightarrow y \in M^\perp$

$$(y, x) = 0 \quad \forall x \in M \quad ?? \quad \text{e' perche'}$$

$$(y_m, x) = 0 \quad \forall m \in \mathbb{N} \quad \forall x \in M$$

$$(y, x) \equiv \lim_m \underbrace{(y_m, x)}_0$$

$$| \underbrace{(y, x) - (y_m, x)}_0 | \xrightarrow{0} 0$$

$$| \underbrace{(y - y_m, x)}_{CS} |$$

$$\|y - y_m\| \|x\|$$

$$\downarrow$$
$$0$$

Teorema delle proiezioni

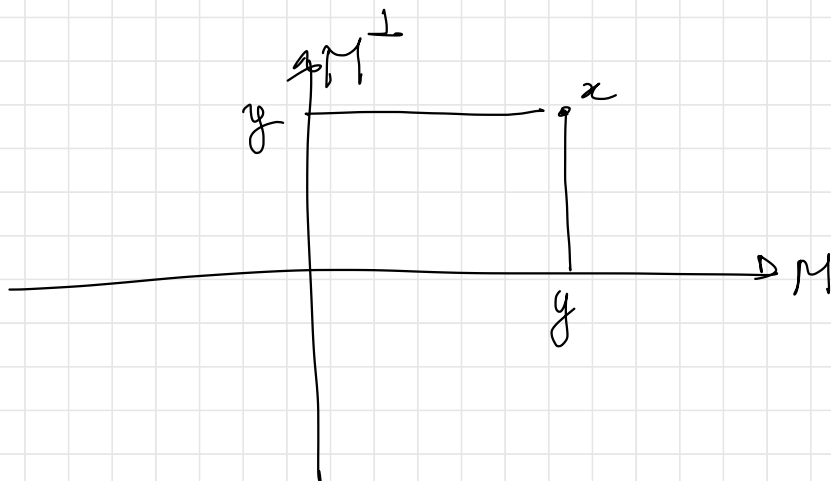
Sia H Hilbert, e sia M sottospazio chiuso.

Allora $\forall x \in H$, esiste un'unica rappresentazione
(scomposizione)
di x come

$$x = \underbrace{y}_{P_M(x)} + \underbrace{z}_{P_{M^\perp}(x)} \quad \text{con } y \in M, \quad z \in M^\perp$$

Inoltre, le applicazioni $x \mapsto y = P_M(x)$
 $x \mapsto z = P_{M^\perp}(x)$

sono op. lineari: $H \rightarrow H$ limitati d'norma 1,
e vale $\|x\|^2 \stackrel{(*)}{=} \|y\|^2 + \|z\|^2$.



dim. • $x = y + z$ $y \perp z$ $(y, z) = 0$

$$\Rightarrow \|y+z\|^2 = (y+z, y+z) = \|y\|^2 + \|z\|^2 + 2(y, z)$$

• Esiste una scomposizione

$$y := P_M(x) \quad z := x - y = x - P_M(x)$$

verifico che $z \in M^\perp$ (avrei $x = \underbrace{y}_M + \underbrace{z}_{M^\perp}$)

$(x - P_M(x)) \in M^\perp$ (?) vero per 2) nel Corollario

• Unicità:

$$x = \underbrace{y_1}_M + \underbrace{z_1}_{M^\perp} \stackrel{(\circledast)}{=} \underbrace{y_2}_M + \underbrace{z_2}_{M^\perp} \Rightarrow \begin{cases} y_1 = y_2 \\ z_1 = z_2 \end{cases}$$

$$\underbrace{y_1 - y_2}_M \stackrel{(\circledast)}{=} \underbrace{z_2 - z_1}_{M^\perp}$$

$$M \cap M^\perp = \{0\}$$

$$x \in M \cap M^\perp \Rightarrow (x, x) = 0 \Rightarrow x = 0$$

- $$\begin{cases} P_M(x_1 + x_2) = P_M(x_1) + P_M(x_2) \\ P_M(\lambda x) = \lambda P_M(x) \end{cases}$$
 linéarité de P_M

$$\begin{aligned} x_1 &= y_1 + z_1 & y_1 &= P_M(x_1) & z_1 &= P_{M^\perp}(x_1) \\ x_2 &= y_2 + z_2 & y_2 &= P_M(x_2) & z_2 &= P_{M^\perp}(x_2) \end{aligned}$$

$$\begin{aligned} x_1 + x_2 &= (y_1 + z_1) + (y_2 + z_2) \\ &= \underbrace{(y_1 + y_2)}_{\in M} + \underbrace{(z_1 + z_2)}_{\in M^\perp} \end{aligned}$$

$$\Rightarrow \underbrace{y_1}_{P_M(x_1)} + \underbrace{y_2}_{P_M(x_2)} = P_M(x_1 + x_2)$$

- $$\|P_M(x)\| \leq \underbrace{M}_1 \|x\|$$

$$\begin{aligned} \|x\|^2 &= \|y\|^2 + \|z\|^2 = \|P_M(x)\|^2 \\ &\quad + \underbrace{\|P_{M^\perp}(x)\|^2}_{\geq 0} \\ \Rightarrow \|P_M(x)\|^2 &\leq \|x\|^2 \end{aligned}$$

- $x \in M \quad x = P_M(x)$

□

Problema: dato H Hilbert, caratterizzare

$H' = \{ \varphi: H \rightarrow \mathbb{R} \text{ lineari e continue} \}$
 $\uparrow$
spazio duale di H

$$H = \mathbb{R}^m$$

$\varphi: H = \mathbb{R}^m \rightarrow \mathbb{R}$ lineare (continua)

$$\varphi: \mathbb{R}^m \rightarrow \mathbb{R}^m$$

$$\varphi(x) = A \cdot x$$

$$m=1 \quad \varphi_a(x) = a \cdot x$$

$\uparrow$
prodotto scalare in $\mathbb{R}^n$

$$\| \varphi_a \|_{(\mathbb{R}^m)} = \| a \|$$

$$| a \cdot x | \leq M \| x \| \quad \text{CS}$$

Sia H Hilbert

Fisso $a \in H$

$$H \ni x \xrightarrow{\varphi_a} (a, x) \in \mathbb{R}$$

Operazio $\varphi_a \in H'$

$$|(a, x)| \leq \|a\| \|x\|$$

$$\|\varphi_a\|_{H'} = \|a\|_H \quad \text{diventa} = \text{prendendo } x=a$$

$$\left| \begin{array}{l} H \ni a \rightsquigarrow \varphi_a \in H' \\ (\text{e ho } \|\varphi_a\|_{H'} = \|a\|_H) \end{array} \right|$$

$$H \subseteq H'$$

Teorema rappresentazione Riesz

Sia H Hilbert, e sia $\varphi \in H'$.

Allora $\exists$ unico $a \in H$ tale che $\varphi = \varphi_a$

ovvero:

$$\varphi(x) = \underset{\uparrow}{(a, x)} \quad \forall x \in H$$

" a rappresenta φ tramite prodotto scalare"

Inoltre

$$\|\varphi\|_{H'} \stackrel{||}{=} \|a\|_H$$
$$\sup_{\substack{x \in H \\ x \neq 0}} \frac{|\varphi(x)|}{\|x\|}$$

(perché $|(a, x)| \leq \|a\| \|x\|$)
con = prendendo $x = a$

$$H \ni a \quad \rightsquigarrow \quad \varphi = \varphi_a \in H'$$

Esempi

1) $H = \mathbb{R}^n$

$$(\mathbb{R}^n)' \quad x \mapsto \overset{f}{\underset{a}{x}}$$

2) $H = L^2(\Omega)$

$$(L^2(\Omega))' \quad g \mapsto (f, g) = \int_{\Omega} f g$$

3) $H = H^1(\Omega)$

$$(H^1(\Omega))' \quad g \mapsto (f, g) = \int_{\Omega} (fg + \nabla f \cdot \nabla g)$$

$$\text{" } H = H' \text{ "}$$